

SUMMER INTERNSHIP REPORT

On

**Satellite Image Land-Use
Classification and Temporal
Change Detection Using Deep
Learning**

Submitted to:



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Amity University Noida

By

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Abstract

This project presents a deep learning framework for satellite image land-use classification and temporal change detection using the EuroSAT dataset. A baseline CNN and a transfer learning model based on ResNet18 were developed and evaluated. The proposed system reuses deep feature embeddings extracted from the trained classifier to perform embedding-based temporal change detection using cosine similarity. Experimental results achieved

- a validation accuracy of 97.06%
- a Macro F1-score of 0.971, and
- a change detection AUC of 0.9769.

In addition to the mandatory deliverables, advanced analyses including GradCAM explainability, multi-threshold change detection, t-SNE embedding visualization, and class imbalance experiments were implemented. A Streamlit-based dashboard was also developed for real-time classification and change detection.

Introduction

Satellite imagery plays a critical role in applications such as environmental monitoring, urban planning, agriculture, and disaster management. Automated analysis of satellite data enables large-scale monitoring while reducing manual effort and improving decision-making. In this project, deep learning techniques were applied to perform land-use classification and temporal change detection. A baseline CNN was first implemented and compared against a transfer learning approach based on ResNet18. The trained model was further reused as a feature extractor for embedding-based change detection. Additional experiments including model explainability, embedding visualization, threshold analysis, and class imbalance evaluation were conducted to improve interpretability and robustness.

Project Deliverables

Deliverable	Status
Data Pipeline & Preprocessing	✓
Baseline CNN	✓
ResNet18 Transfer Learning	✓
Confusion Matrix & F1 Analysis	✓
ROC-based Change Detection	✓
Change Heatmaps	✓
Error Analysis	✓
Streamlit Dashboard	✓
GradCAM Explainability	✓ Bonus
Multi-Threshold Analysis	✓ Bonus
t-SNE Embedding Visualization	✓ Bonus
Class Imbalance Experiment	✓ Bonus

Dataset Description

- **EuroSAT RGB Dataset**

The EuroSAT RGB dataset contains approximately 27,000 satellite images distributed across 10 land-use classes: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. Images are captured by the Sentinel-2 Earth observation mission and serve as the primary dataset for model training and evaluation.

- **UC Merced Land Use Dataset**

The UC Merced Land Use dataset contains 2,100 aerial images across 21 land-use categories. It is used as an external holdout dataset to evaluate the generalization capability of the trained model.

Data Preprocessing

All images were resized to 224×224 pixels, converted to tensors, and normalized using ImageNet mean and standard deviation values. The EuroSAT dataset was divided into 70% training, 15% validation, and 15% testing subsets. Data augmentation techniques including random horizontal flips and rotations were applied during training to improve model robustness.

Methodology Overview

The proposed framework consists of three modules:

1. Land-Use Classification – A baseline CNN and a transfer learning model based on ResNet18 were trained for satellite image classification.
2. Temporal Change Detection – Feature embeddings extracted from the trained ResNet18 backbone were compared using cosine similarity to identify temporal changes between image pairs.
3. Geo-Dashboard – A Streamlit application was developed to classify uploaded satellite images, compute similarity scores, visualize change heatmaps, and support multiple threshold operating modes.

Workflow:

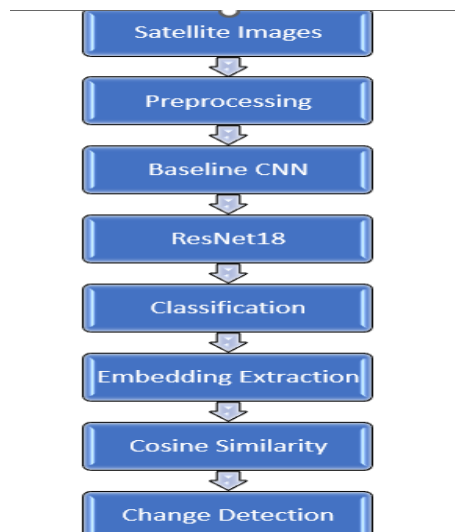


Figure 1. Overall workflow of the proposed satellite image classification and temporal change detection framework.

Baseline CNN Model

A 3-layer Convolutional Neural Network (CNN) was implemented as a baseline model for satellite image classification. The architecture consisted of three convolutional layers with ReLU activation and max-pooling, followed by fully connected layers. The model was trained on the EuroSAT training set using cross-entropy loss and the Adam optimizer.

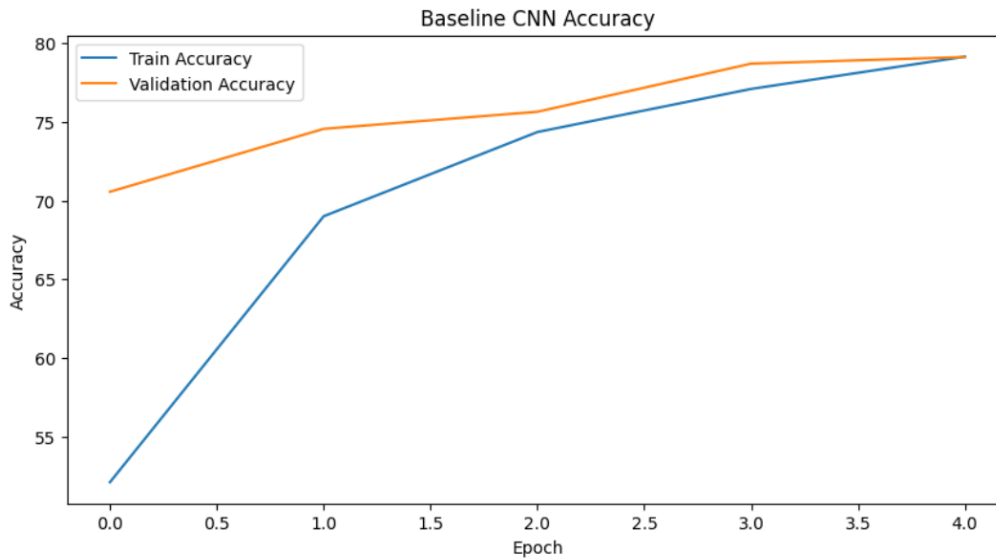


Figure 2. Training and validation accuracy curves for the baseline CNN model.

Results

Metric	Value
Validation Accuracy	79.14%
Training Epochs	5
Optimizer	Adam
Loss Function	Cross Entropy

The baseline CNN established a performance benchmark for comparison with transfer learning approaches.

Observations

The baseline model demonstrated the feasibility of satellite image classification using deep learning; however, further performance improvements were required for practical deployment. This motivated the use of transfer learning with ResNet18 in the next phase of the project.

Transfer Learning Using ResNet18

To improve classification performance, transfer learning was implemented using the ResNet18 architecture pre-trained on ImageNet. Transfer learning enables the model to leverage previously learned visual features and adapt them to satellite image classification.

The training process was performed in two phases. In Phase 1, the pre-trained convolutional layers were frozen and only the final classification layer was trained. In Phase 2, the last residual blocks were unfrozen and fine-tuned using a lower learning rate to improve feature adaptation to satellite imagery.

Results

The transfer learning approach significantly outperformed the baseline CNN model. Validation accuracy increased from 79.14% to 97.06%, demonstrating the effectiveness of deep pre-trained feature extraction.

Model	Validation Accuracy
Baseline CNN	79.14%
ResNet18 (Phase 1)	90.86%
ResNet18 (Phase 2)	97.06%

The training and validation accuracy curves for ResNet18 are shown in Figure 2.

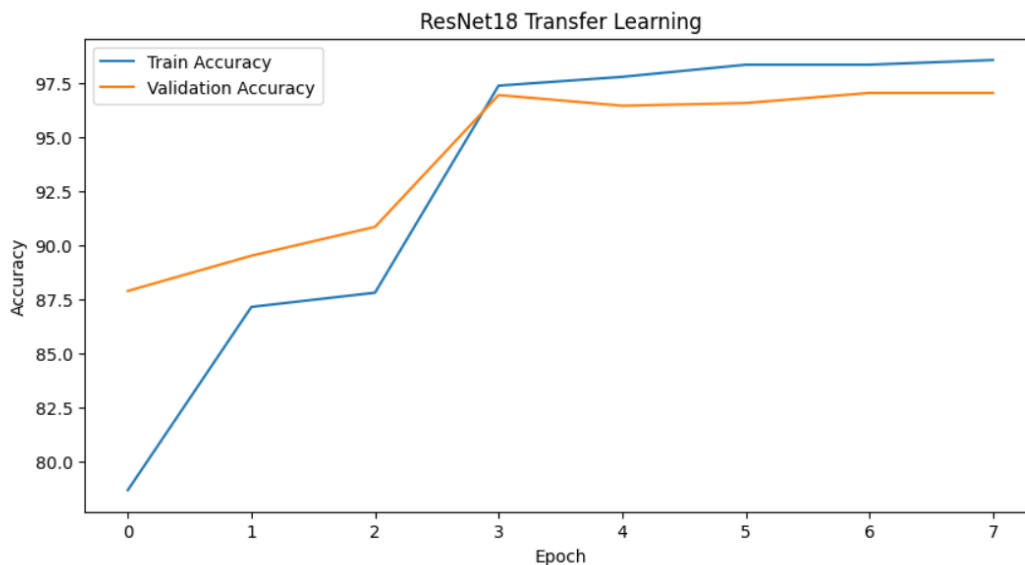


Figure 3. Training and validation accuracy curves for ResNet18 transfer learning.

ResNet18 Final Performance

Metric	Value
Validation Accuracy	97.06%
Macro F1 Score	0.971
Change Detection AUC	0.9769
Backbone	ResNet18

Model Evaluation and Performance Analysis

The performance of the fine-tuned ResNet18 model was evaluated using accuracy, F1-score, confusion matrix analysis, and ROC-based change detection metrics.

Classification Performance

The fine-tuned ResNet18 model achieved a validation accuracy of 97.06%, a Macro F1-score of 0.971, and a change detection AUC of 0.9769, demonstrating strong classification and temporal change detection performance.

Metric	Value
Validation Accuracy	97.06%
Macro F1 Score	0.971
Change Detection AUC	0.9769

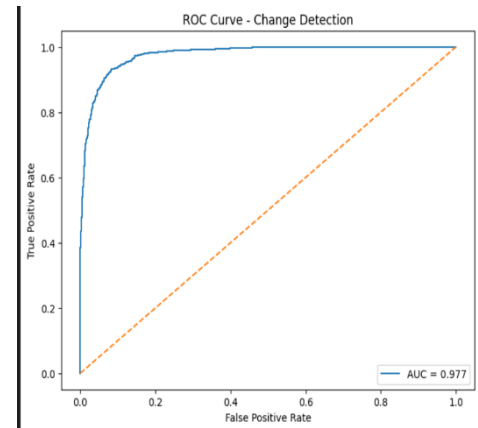
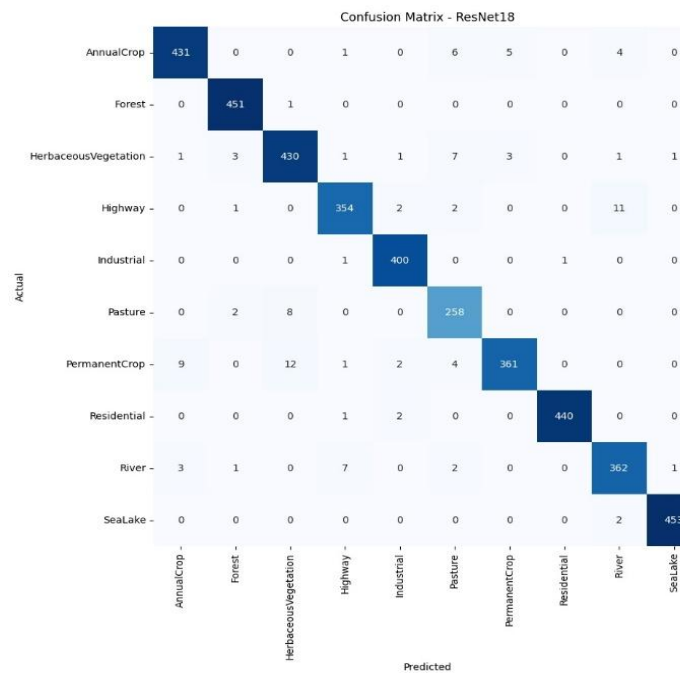


Figure 4. Confusion Matrix for ResNet18 Classification

Figure 5. Roc curve

The confusion matrix indicates strong class-wise classification performance with minimal confusion among visually similar categories. The ROC curve demonstrates excellent discrimination capability for change detection, achieving an AUC of 0.9769.

Evaluation on the UC Merced holdout dataset demonstrated strong generalization performance, confirming the effectiveness of transfer learning for unseen satellite imagery.

Temporal Change Detection and Error Analysis

The trained ResNet18 backbone was reused as a feature extractor to generate 512-dimensional image embeddings. Cosine similarity was computed between image pairs, and a similarity threshold of 0.6523 was selected using ROC analysis. Image pairs below the threshold were classified as changed. Heatmaps were generated to visualize spatial differences between

corresponding image pairs.



Figure 6. Example No-Change Detection



Figure 7. Example Change Detection

Error Analysis

The top five misclassified samples were analyzed to identify failure patterns. Most errors occurred between visually similar land-use categories such as vegetation-related classes and transportation-related classes. These results indicate that classification errors were primarily caused by inter-class visual similarity rather than model instability.



Figure 8. Top Five Misclassified Samples

Bonus A: GradCAM Visualization

GradCAM was applied to the fine-tuned ResNet18 model to identify image regions responsible for classification decisions. Visualizations confirmed that the model focused on meaningful land-cover structures such as vegetation patterns, water bodies, and urban regions.

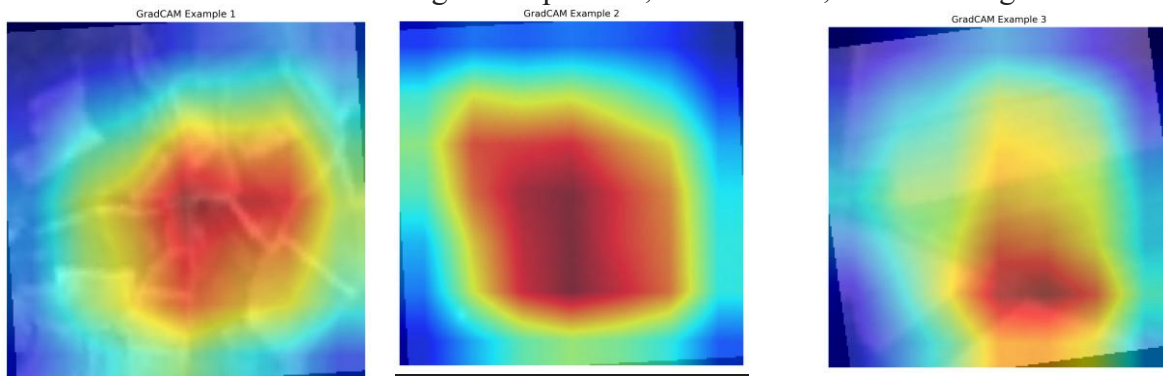


Figure 9. Gradcam images

Bonus B: Multi-Threshold Change Detection

Three operating modes were implemented: High Recall, Balanced, and High Precision. The

dashboard allows users to adjust the similarity threshold and observe changes in detection sensitivity.

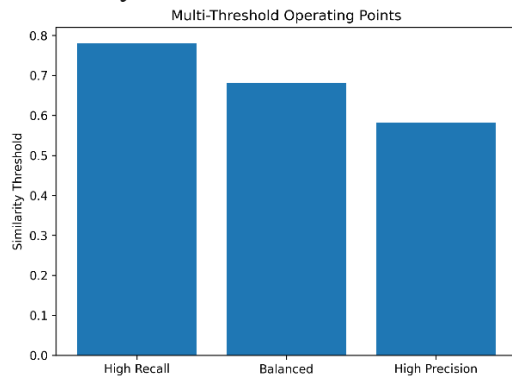


Figure 10. Comparison of High Recall, Balanced, and High Precision operating thresholds.

Bonus C: Embedding Visualization

Feature embeddings extracted from ResNet18 were projected into two dimensions using t-SNE. Distinct class clusters demonstrate that the model learned discriminative feature representations.

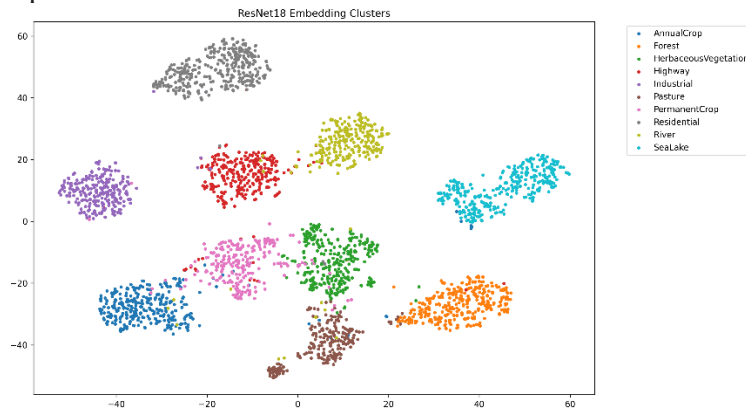


Figure 11. Class-wise t-SNE visualization of ResNet18 feature embeddings.

Bonus D: Imbalance Experiment

Two classes were downsampled to 20% of their original size. Weighted loss was subsequently applied to mitigate class imbalance and partially recover performance.

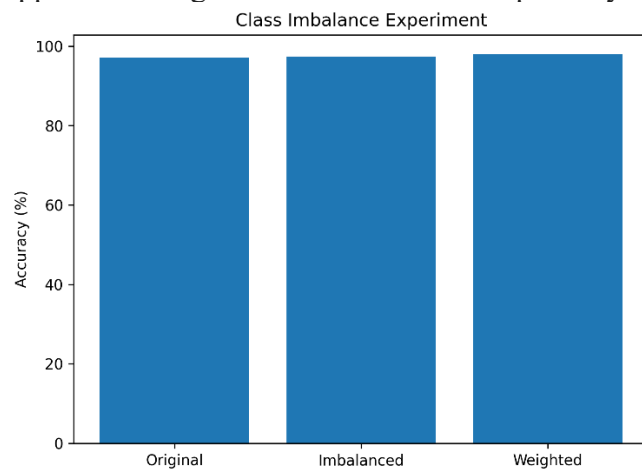


Figure 12. Performance comparison under class imbalance and weighted-loss mitigation.

Streamlit Dashboard

An interactive Streamlit dashboard was developed to provide real-time land-use classification and temporal change detection. The application accepts two satellite images representing different time periods and generates the following outputs:

- Predicted land-use class for each image
- Classification confidence score
- Cosine similarity score between image embeddings
- Change/No Change decision based on the selected threshold
- Visual change heatmap highlighting differences between image pairs

The dashboard operates entirely offline after setup and enables intuitive visualization of classification and change detection results.

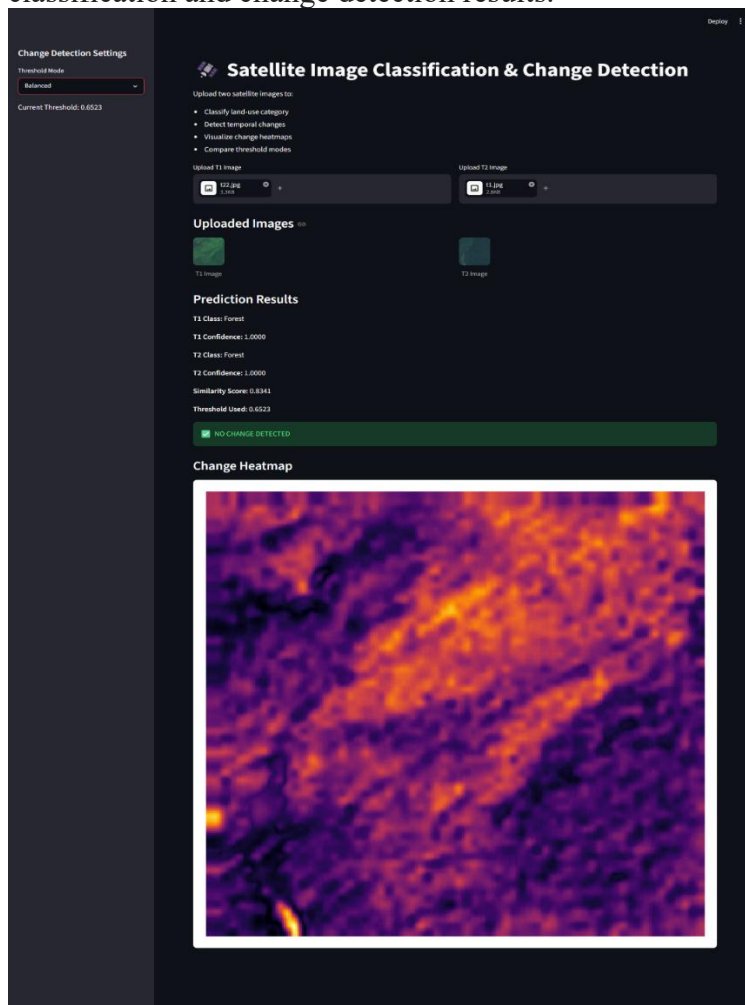


Figure 13. Dashboard

Conclusion

This project successfully developed a deep learning framework for satellite image land-use classification and temporal change detection. A baseline CNN and a transfer learning model based on ResNet18 were implemented and evaluated on the EuroSAT dataset.

The final ResNet18 model achieved a validation accuracy of 97.06% and a Macro F1-score of 0.971. For temporal change detection, cosine similarity between feature embeddings was used to identify land-use changes, achieving an AUC score of 0.9769. An interactive Streamlit dashboard was also deployed to provide user-friendly visualization and inference capabilities. The results demonstrate the effectiveness of transfer learning and embedding-based similarity analysis for remote sensing applications.